

IS311 Web Development

Week 5 - Lecture 6

Dr. Abbas Malik
amaalik@psu.edu.sa

Bootstrap

- HTML, CSS and Javascript Front End Framework
- Responsive and mobile-first websites
- Faster and easier web development
- Includes HTML & CSS design templates
 - Typography, forms, buttons, tables, navigation, modals, image carousels, ...

Bootstrap

- Optional JavaScript plugins
- Not supported for IE11 and earlier.

Responsive web design is about creating web sites which **automatically adjust** themselves to look good on **all devices, from small phones to large desktops.**

Bootstrap History

- Twitter Blueprint - Mark Otto & Jacob Thornton at Twitter (Bootstrap 1)
- Bootstrap 2 - Jan 2012
- Bootstrap 3 - Aug 2013
- Bootstrap 4 - Oct 2014
- Bootstrap 5 - May 2021

Bootstrap

- CSS is written in **Sass** (Syntactically Awesome Style Sheets) - <https://sass-lang.com/>
 - Compiled CSS files are available for download or direct include from CDN
<https://www.jsdelivr.com/>
- **JavaScript**

Download from <https://getbootstrap.com/>

How to use Bootstrap 5

- Directly link from CDN website

```
<!-- Latest compiled and minified CSS -->  
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"  
      rel="stylesheet">  
  
<!-- Latest compiled JavaScript -->  
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.m  
in.js"></script>
```

Containers

- Requires a **containing element** to wrap site contents
- **.container** class - fixed width container
- **.container-fluid** class - a full width container



Fixed Container - **.container**

- Responsive, fixed-width container>

		Small ≥576px	Medium ≥768px	Large ≥992px	Extra Large ≥1200px	XXL ≥1400px
max-width	100%	540px	720px	960px	1140px	1320px

```
<div class="container">
  <h1>My First Bootstrap Page</h1>
  <p>This is some text.</p>
</div>
```

My First Bootstrap Page

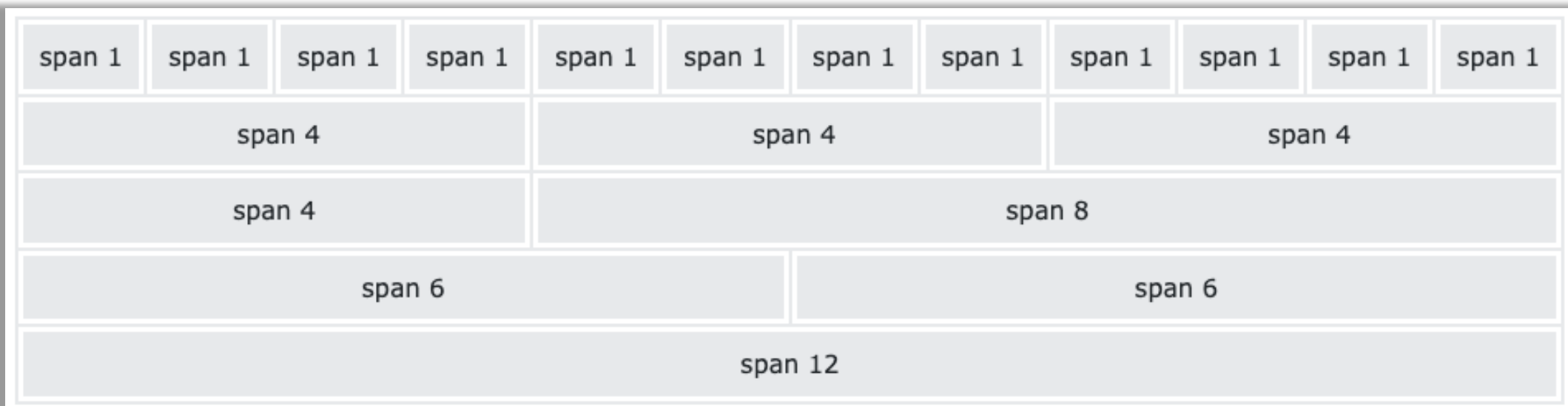
This part is inside a .container class.

The .container class provides a responsive fixed width container.

Resize the browser window to see that the container width will change at different breakpoints.

5 Grid System

- Built with **flexbox**
- Allows up to 12 columns across the page
- Wider Columns by grouping

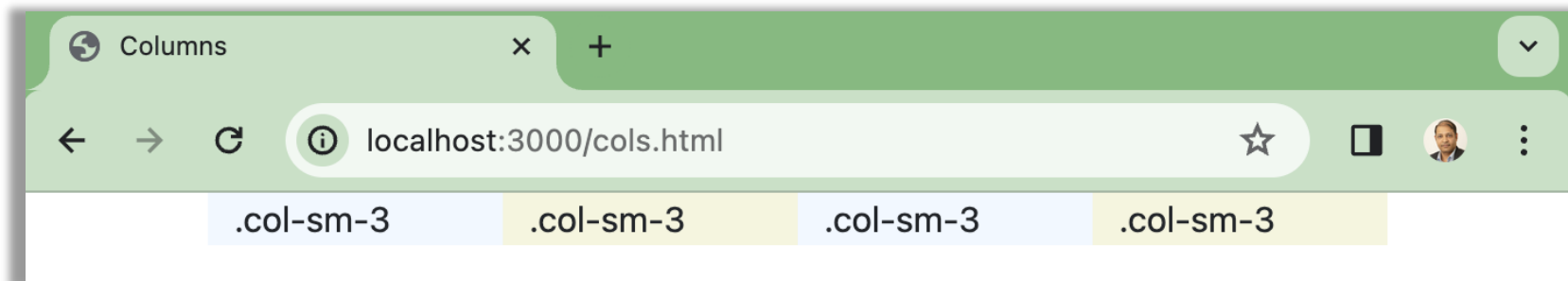


Grid Classes

- **.col-** extra small devices ($\leq 576\text{px}$)
- **.col-sm-** small devices ($\geq 576\text{px}$)
- **.col-md-** medium devices ($\geq 768\text{px}$)
- **.col-lg-** large devices ($\geq 992\text{px}$)
- **.col-xl-** xlarge devices ($\geq 1200\text{px}$)
- **.col-xxl-** xxlarge devices ($\geq 1400\text{px}$)

Grid Equal

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
    rel="stylesheet">
  <title>Columns</title>
</head>
<body>
  <div class="container">
    <div class="row">
      <div class="col-sm-3" style="background-color: #add8e6;">.col-sm-3</div>
      <div class="col-sm-3" style="background-color: #f5deb3;">.col-sm-3</div>
      <div class="col-sm-3" style="background-color: #add8e6;">.col-sm-3</div>
      <div class="col-sm-3" style="background-color: #f5deb3;">.col-sm-3</div>
    </div>
  </div>
</body>
</html>
```



```

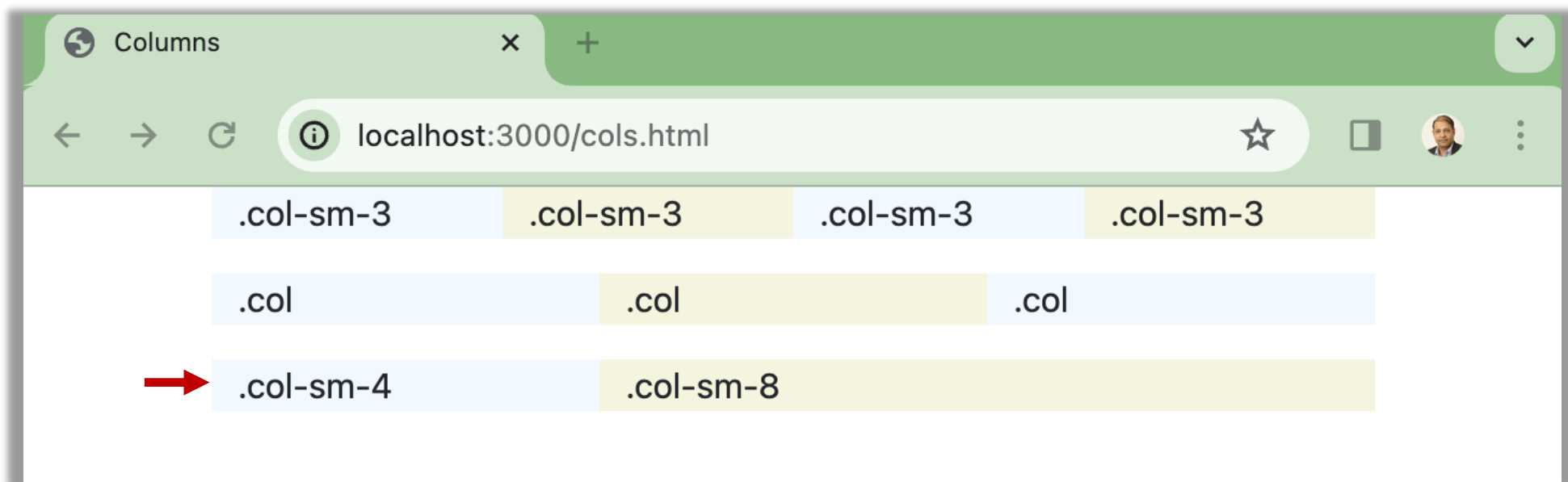
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
    rel="stylesheet">
  <title>Columns</title>
</head>
<body>
  <div class="container">
    <div class="row">
      <div class="col-sm-3" style="background-color: #aliceblue;">.col-sm-3</div>
      <div class="col-sm-3" style="background-color: #beige;">.col-sm-3</div>
      <div class="col-sm-3" style="background-color: #aliceblue;">.col-sm-3</div>
      <div class="col-sm-3" style="background-color: #beige;">.col-sm-3</div>
    </div>
  </div>
</body>
</html>

```



Unequal Grid

```
<div class="row">
<div class="col-sm-4" style="background-color: aliceblue;">.col-sm-4</div>
<div class="col-sm-8" style="background-color: beige;">.col-sm-8</div>
</div>
```



Responsive and Nonresponsive Grids

The image shows a code editor on the left and a web browser on the right. The code editor displays the HTML code for a responsive grid layout using Bootstrap classes. The browser shows the rendered output of this code, demonstrating how the grid adapts to different screen widths.

Code Editor (cols.html):

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
7     rel="stylesheet">
8   <title>Columns</title>
9 </head>
10 <body>
11   <div class="container">
12     <div class="row">
13       <div class="col-sm-3" style="background-color: #aliceblue;">.col-sm-3</div>
14       <div class="col-sm-3" style="background-color: #beige;">.col-sm-3</div>
15       <div class="col-sm-3" style="background-color: #aliceblue;">.col-sm-3</div>
16       <div class="col-sm-3" style="background-color: #beige;">.col-sm-3</div>
17     </div>
18     <p></p>
19     <div class="row">
20       <div class="col" style="background-color: #aliceblue;">.col</div>
21       <div class="col" style="background-color: #beige;">.col</div>
22       <div class="col" style="background-color: #aliceblue;">.col</div>
23     </div>
24     <p></p>
25     <div class="row">
26       <div class="col-sm-4" style="background-color: #aliceblue;">.col-sm-4</div>
27       <div class="col-sm-8" style="background-color: #beige;">.col-sm-8</div>
28     </div>
29   </div>
30 </body>
31 </html>
32
33

```

Browser (localhost:3000/cols.html):

The browser displays the rendered grid. It shows three rows of columns. The first row has four columns of width .col-sm-3. The second row has three columns of width .col. The third row has two columns of width .col-sm-4 and .col-sm-8. The columns are styled with alternating background colors: #aliceblue and #beige.

Typography/Text

- **font-size** of 1rem (16px)
- **line-height** is 1.5
- **<p>**
 - **margin-top**: 0
 - **margin-bottom**: 1rem (16px)
- **<h1> - <h6>**: bold and responsive font-size

h1 Bootstrap heading

h2 Bootstrap heading

h3 Bootstrap heading

h4 Bootstrap heading

h5 Bootstrap heading

h6 Bootstrap heading

Heading classes

```
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</head>
<body>

<div class="container mt-3">
  <p class="h1">h1 Bootstrap heading</p>
  <p class="h2">h2 Bootstrap heading</p>
  <p class="h3">h3 Bootstrap heading</p>
  <p class="h4">h4 Bootstrap heading</p>
  <p class="h5">h5 Bootstrap heading</p>
  <p class="h6">h6 Bootstrap heading</p>
</div>

</body>
</html>
```

h1 Bootstrap heading

h2 Bootstrap heading

h3 Bootstrap heading

h4 Bootstrap heading

h5 Bootstrap heading

h6 Bootstrap heading

<small> in <h#>

```
</head>
<body>

<div class="container mt-3">
  <h1>Smaller, Secondary Text</h1>
  <p>The small element (and the .small class) is used to create a smaller, secondary text in any heading:</p>
  <h1>h1 heading <small>secondary text</small></h1>
  <h2>h2 heading <small>secondary text</small></h2>
  <h3>h3 heading <small>secondary text</small></h3>
  <h4>h4 heading <small>secondary text</small></h4>
  <h5>h5 heading <small>secondary text</small></h5>
  <h6>h6 heading <small>secondary text</small></h6>
</div>

</body>
</html>
```

Smaller, Secondary Text

The small element (and the .small class) is used to create a smaller, secondary text in any heading:

h1 heading secondary text

h2 heading secondary text

h3 heading secondary text

h4 heading secondary text

h5 heading secondary text

h6 heading secondary text

Display Headings

```
<div class="container mt-3">
  <h1>Display Headings</h1>
  <p>Display headings are used to stand out more than normal headings (larger font-size and lighter font-weight):</p>
  <h1 class="display-1">Display 1</h1>
  <h1 class="display-2">Display 2</h1>
  <h1 class="display-3">Display 3</h1>
  <h1 class="display-4">Display 4</h1>
  <h1 class="display-5">Display 5</h1>
  <h1 class="display-6">Display 6</h1>
</div>
```

Display Headings

Display headings are used to stand out more than normal headings (larger font-size and lighter font-weight):

Display 1

Display 2

Display 3

Display 4

Display 5

Text Colors

- Contextual classes - provides meaning through colors

.text-muted

.text-primary

.text-success

.text-info

.text-warning

.text-secondary

.text-white

.text-dark

.text-body

.text-light

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js">
</script>
</head>
<body>

<div class="container mt-3">
  <h2>Contextual Colors</h2>
  <p>Use the contextual classes to provide "meaning through colors":</p>
  <p class="text-muted">This text is muted.</p>
  <p class="text-primary">This text is important.</p>
  <p class="text-success">This text indicates success.</p>
  <p class="text-info">This text represents some information.</p>
  <p class="text-warning">This text represents a warning.</p>
  <p class="text-danger">This text represents danger.</p>
  <p class="text-secondary">Secondary text.</p>
  <p class="text-dark">This text is dark grey.</p>
  <p class="text-body">Default body color (often black).</p>
  <p class="text-light">This text is light grey (on white background).</p>
  <p class="text-white">This text is white (on white background).</p>
</div>

</body>
</html>
```

Contextual Colors

Use the contextual classes to provide "meaning through colors":

This text is muted.

This text is important.

This text indicates success.

This text represents some information.

This text represents a warning.

This text represents danger.

Secondary text.

This text is dark grey.

Default body color (often black).

This text is light grey (on white background).

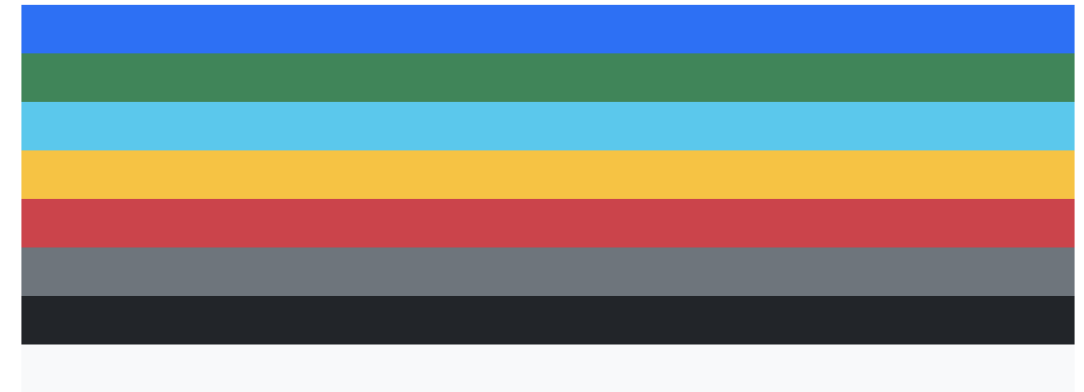
```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js">
</script>
</head>
<body>

<div class="container mt-3">
  <h2>Contextual Backgrounds</h2>
  <p>Use the contextual background classes to provide "meaning through colors".</p>
  <div class="bg-primary p-3"></div>
  <div class="bg-success p-3"></div>
  <div class="bg-info p-3"></div>
  <div class="bg-warning p-3"></div>
  <div class="bg-danger p-3"></div>
  <div class="bg-secondary p-3"></div>
  <div class="bg-dark p-3"></div>
  <div class="bg-light p-3"></div>
</div>

</body>
</html>
```

Contextual Backgrounds

Use the contextual background classes to provide "meaning through colors".



Table

```

</script>
</head>
<body>

<div class="container mt-3">
  <h2>Basic Table</h2>
  <p>The .table class adds basic styling (light padding and horizontal dividers) to a
table:</p>
  <table class="table">
    <thead>
      <tr>
        <th>Firstname</th>
        <th>Lastname</th>
        <th>Email</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>John</td>
        <td>Doe</td>
        <td>john@example.com</td>
      </tr>
      <tr>
        <td>Mary</td>
        <td>Moe</td>
        <td>mary@example.com</td>
      </tr>
      <tr>
        <td>July</td>
        <td>Dooley</td>
        <td>july@example.com</td>
      </tr>
    </tbody>
  </table>
</div>

</body>
</html>

```

Basic Table

The .table class adds basic styling (light padding and horizontal dividers) to a table:

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

Striped Rows

`.table-striped`

```
<table class="table table-striped">
```

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

Bordered Table

`.table-bordered`

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

Hover Rows

`.table-hover`

Hover Rows

The `.table-hover` class enables a hover state (grey background on mouse over) on table rows:

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

Black/Dark Table `.table-dark`

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

Dark Striped Table

```
<table class="table table-dark table-striped">
```

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

BorderlessTable

`.table-borderless`

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

Contextual Classes

```
<tbody>
  <tr>
    <td>Default</td>
    <td>Defaultson</td>
    <td>def@somemail.com</td>
  </tr>
  <tr class="table-primary">
    <td>Primary</td>
    <td>Joe</td>
    <td>joe@example.com</td>
  </tr>
  <tr class="table-success">
    <td>Success</td>
    <td>Doe</td>
    <td>john@example.com</td>
  </tr>
  <tr class="table-danger">
    <td>Danger</td>
    <td>Moe</td>
    <td>mary@example.com</td>
  </tr>
  <tr class="table-info">
    <td>Info</td>
    <td>Dooley</td>
    <td>july@example.com</td>
  </tr>
  <tr class="table-warning">
    <td>Warning</td>
    <td>Refs</td>
    <td>bo@example.com</td>
  </tr>
  <tr class="table-active">
    <td>Active</td>
    <td>Activeson</td>
    <td>act@example.com</td>
  </tr>
  <tr class="table-secondary">
    <td>Secondary</td>
    <td>Secondson</td>
    <td>sec@example.com</td>
  </tr>
  <tr class="table-light">
    <td>Light</td>
    <td>Angie</td>
    <td>angie@example.com</td>
  </tr>
  <tr class="table-dark">
    <td>Dark</td>
    <td>Bo</td>
    <td>bo@example.com</td>
  </tr>
</tbody>
```

Firstname	Lastname	Email
Default	Defaultson	def@somemail.com
Primary	Joe	joe@example.com
Success	Doe	john@example.com
Danger	Moe	mary@example.com
Info	Dooley	july@example.com
Warning	Refs	bo@example.com
Active	Activeson	act@example.com
Secondary	Secondson	sec@example.com
Light	Angie	angie@example.com
Dark	Bo	bo@example.com

Contextual Classes

Class	Description
<code>.table-primary</code>	Blue: Indicates an important action
<code>.table-success</code>	Green: Indicates a successful or positive action
<code>.table-danger</code>	Red: Indicates a dangerous or potentially negative action
<code>.table-info</code>	Light blue: Indicates a neutral informative change or action
<code>.table-warning</code>	Orange: Indicates a warning that might need attention
<code>.table-active</code>	Grey: Applies the hover color to the table row or table cell
<code>.table-secondary</code>	Grey: Indicates a slightly less important action
<code>.table-light</code>	Light grey table or table row background
<code>.table-dark</code>	Dark grey table or table row background

Responsive Table

```
<div class="table-responsive">
  <table class="table">
    ...
  </table>
</div>
```

#	Firstname	Lastname	Age	City	Country	Sex	Example	Example	Example	Example	Example	Example	Example	Example
1	Anna	Pitt	35	New York	USA	Female	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Responsive Table

Class	Screen width
<code>.table-responsive-sm</code>	< 576px
<code>.table-responsive-md</code>	< 768px
<code>.table-responsive-lg</code>	< 992px
<code>.table-responsive-xl</code>	< 1200px
<code>.table-responsive-xxl</code>	< 1400px

Images - Rounded Corners

```

```



Images - Rounded Corners

```

```



Images - Rounded Corners

```

```



Image Alignment

```


<p>Paragraph</p>
<p>Paragraph</p>
<p>
```



Paragraph

Paragraph

Paragraph



Image - Center Alignment

```

```




```

< images.html > < html > < body > < div.container > < p > < img.float-end
1  CTYPE html>
2  |l lang="en">
3  d>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-sc
6  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist
7  <title>Images</title>
8  ad>
9  y style="background-color: □aliceblue;">
10 <div class="container">
11   <p>
12     
13     
16   <p>
17     
18     
19   </p>
20   <p>Paragraph</p>
21   <p>Paragraph</p>
22   <p>Paragraph</p>
23   <p>Paragraph</p>
24   <p>Paragraph</p>
25   <p>Paragraph</p>
26   <p>Paragraph</p>
27   <p>Paragraph</p>
28   
29   text after image
30 </div>
31 dy>
32 ml>

```



text after image

Jumbotron

Jumbotron Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat..

[More details](#)

Alerts

- An easy way to create predefined alert messages

`.alert` `.alert-success` `.alert-info` `.alert-warning`

`.alert-danger` `.alert-primary` `.alert-secondary`

`.alert-light` `.alert-dark`

Alerts

Success! This alert box could indicate a successful or positive action.

Info! This alert box could indicate a neutral informative change or action.

Warning! This alert box could indicate a warning that might need attention.

Danger! This alert box could indicate a dangerous or potentially negative action.

Primary! Indicates an important action.

Secondary! Indicates a slightly less important action.

Dark! Dark grey alert.

[More details](#)

Closing Alerts

- To close the alert message, add a **.alert-dismissible** class to the alert container

```
<div class="alert alert-success alert-dismissible">
  <button type="button" class="btn-close" data-bs-
dismiss="alert"></button>
  <strong>Success!</strong> This alert box could indicate a
successful or positive action.
</div>
```

Success! This alert box could indicate a successful or positive action.



[More details](#)

Buttons

- Different styles of buttons

Basic

Primary

Secondary

Success

Info

Warning

Danger

Dark

Light

[Link](#)

```
<button type="button" class="btn">Basic</button>
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-dark">Dark</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-link">Link</button>
```

Button

- Button classes can be used on `<a>`, `<button>`, or `<input>` elements

```
<a href="#" class="btn btn-success">Link Button</a>
<button class="btn btn-success">Button</button>
<input type="button" class="btn btn-success" value="Input Button">
<input type="submit" class="btn btn-success" value="Submit Button">
<input type="reset" class="btn btn-success" value="Reset Button">
```

Button Elements

Link Button

Button

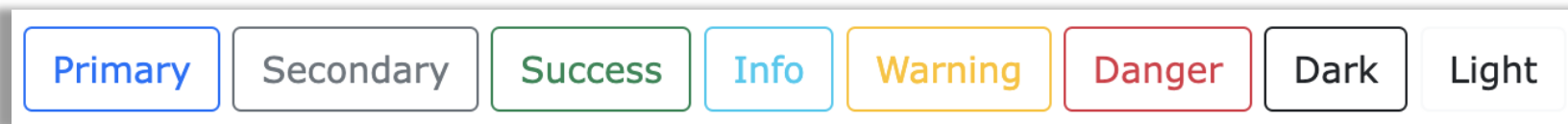
Input Button

Submit Button

Reset Button

Button Outline

```
<button type="button" class="btn btn-outline-primary">Primary</button>
<button type="button" class="btn btn-outline-
secondary">Secondary</button>
<button type="button" class="btn btn-outline-success">Success</button>
<button type="button" class="btn btn-outline-info">Info</button>
<button type="button" class="btn btn-outline-warning">Warning</button>
<button type="button" class="btn btn-outline-danger">Danger</button>
<button type="button" class="btn btn-outline-dark">Dark</button>
<button type="button" class="btn btn-outline-light text-
dark">Light</button>
```



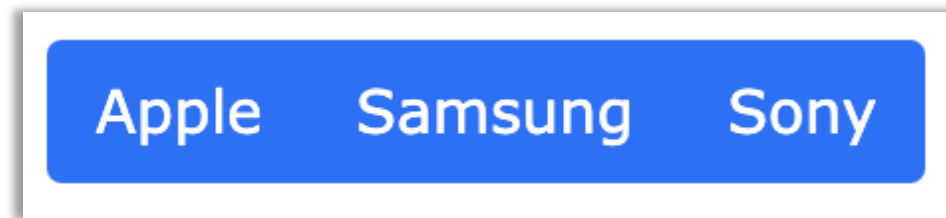
Buttons

- Button Size
- Block Level Buttons
- Active/Disabled Buttons
- Spinner Buttons

[More details](#)

Button Groups

- Group a series of buttons together in a button group
- Display buttons horizontally or vertically without space
- Navigation bar



[More details](#)

[Button group tutorial](#)

Badges

- Used to add additional information to any content

```
<h1>Example heading <span class="badge bg-secondary">New</span></h1>
<h2>Example heading <span class="badge bg-secondary">New</span></h2>
<h3>Example heading <span class="badge bg-secondary">New</span></h3>
<h4>Example heading <span class="badge bg-secondary">New</span></h4>
<h5>Example heading <span class="badge bg-secondary">New</span></h5>
<h6>Example heading <span class="badge bg-secondary">New</span></h6>
```

Example heading **New**

Example heading **New**

Example heading **New**

Example heading **New**

Example heading **New**

Example heading **New**

Contextual Badges

```

<span class="badge bg-primary">Primary</span>
<span class="badge bg-secondary">Secondary</span>
<span class="badge bg-success">Success</span>
<span class="badge bg-danger">Danger</span>
<span class="badge bg-warning">Warning</span>
<span class="badge bg-info">Info</span>
<span class="badge bg-light">Light</span>
<span class="badge bg-dark">Dark</span>
  
```

Primary Secondary Success Danger Warning Info Light Dark

Round Pill Badges

```
<span class="badge rounded-pill bg-primary">Primary</span>
<span class="badge rounded-pill bg-secondary">Secondary</span>
<span class="badge rounded-pill bg-success">Success</span>
<span class="badge rounded-pill bg-danger">Danger</span>
<span class="badge rounded-pill bg-warning">Warning</span>
<span class="badge rounded-pill bg-info">Info</span>
<span class="badge rounded-pill bg-light">Light</span>
<span class="badge rounded-pill bg-dark">Dark</span>
```

Primary

Secondary

Success

Danger

Warning

Info

Light

Dark

Badge inside an Element

```

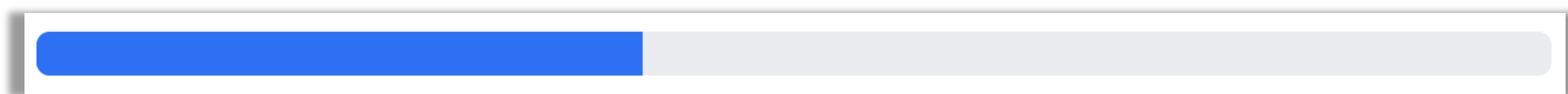
<button type="button" class="btn btn-primary">
  Messages <span class="badge bg-danger">4</span>
</button>

```

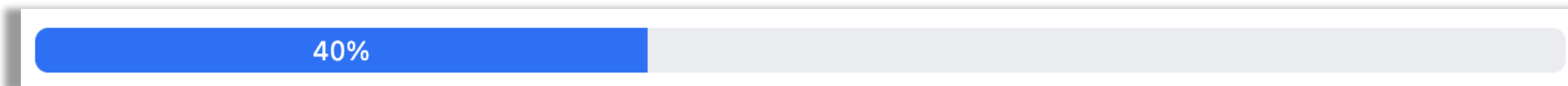


Progress Bar

```
<div class="progress" style="height:20px">
  <div class="progress-bar" style="width:40%;"></div>
</div>
```

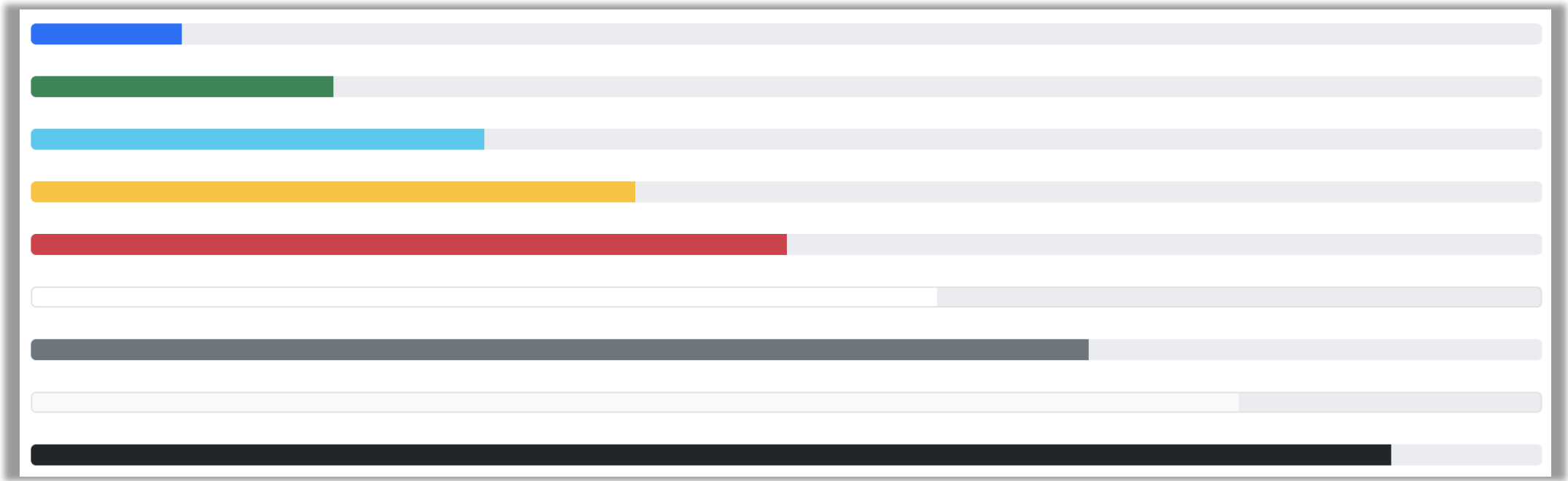


```
<div class="progress" style="height:20px">
  <div class="progress-bar" style="width:40%;">40%</div>
</div>
```



Colored Progress Bars

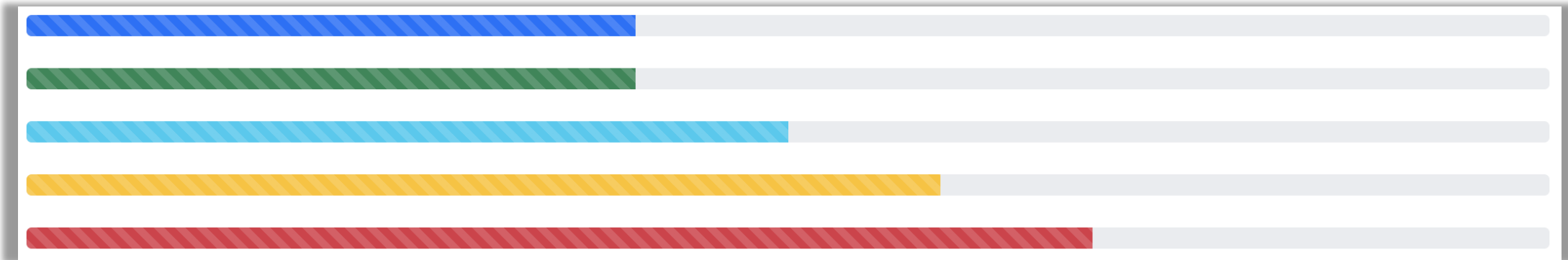
- Using **bg-*** can change colors of progress bar



Striped Progress Bars

.progress-bar-striped

```
<div class="progress">
  <div class="progress-bar progress-bar-striped" style="width:40%"></div>
</div>
```

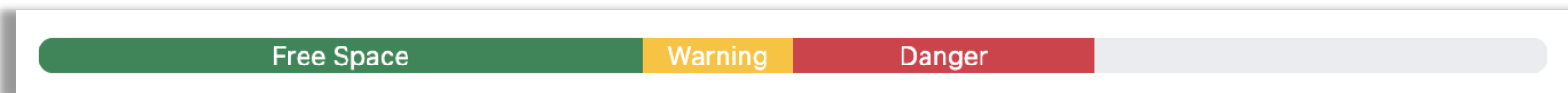


.progress-bar-animated

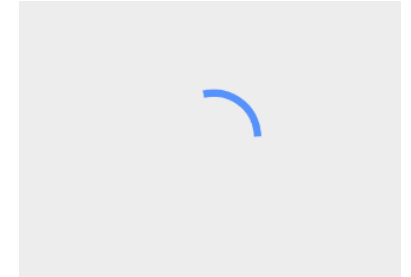
```
<div class="progress-bar progress-bar-striped progress-bar-animated"
style="width:40%"></div>
```

Multiple Progress Bars

```
<div class="progress">
  <div class="progress-bar bg-success" style="width:40%">
    Free Space
  </div>
  <div class="progress-bar bg-warning" style="width:10%">
    Warning
  </div>
  <div class="progress-bar bg-danger" style="width:20%">
    Danger
  </div>
</div>
```



Spinner



`.spinner-border`

```
<div class="spinner-border text-muted"></div>
<div class="spinner-border text-primary"></div>
<div class="spinner-border text-success"></div>
<div class="spinner-border text-info"></div>
<div class="spinner-border text-warning"></div>
<div class="spinner-border text-danger"></div>
<div class="spinner-border text-secondary"></div>
<div class="spinner-border text-dark"></div>
<div class="spinner-border text-light"></div>
```


Grow Spinners

.spinner-grow

```
<div class="spinner-grow text-muted"></div>
<div class="spinner-grow text-primary"></div>
<div class="spinner-grow text-success"></div>
<div class="spinner-grow text-info"></div>
<div class="spinner-grow text-warning"></div>
<div class="spinner-grow text-danger"></div>
<div class="spinner-grow text-secondary"></div>
<div class="spinner-grow text-dark"></div>
<div class="spinner-grow text-light"></div>
```

Spinner

- Spinner Size

```
<div class="spinner-border spinner-border-sm"></div>
<div class="spinner-grow spinner-grow-sm"></div>
```

- Spinner Button

```
<button class="btn btn-primary">
  <span class="spinner-border spinner-border-sm"></span>
  Loading..
</button>
```

Pagination

.pagination

.page-item

.page-link

```
<ul class="pagination">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```



Active State

```
<ul class="pagination">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item active"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```



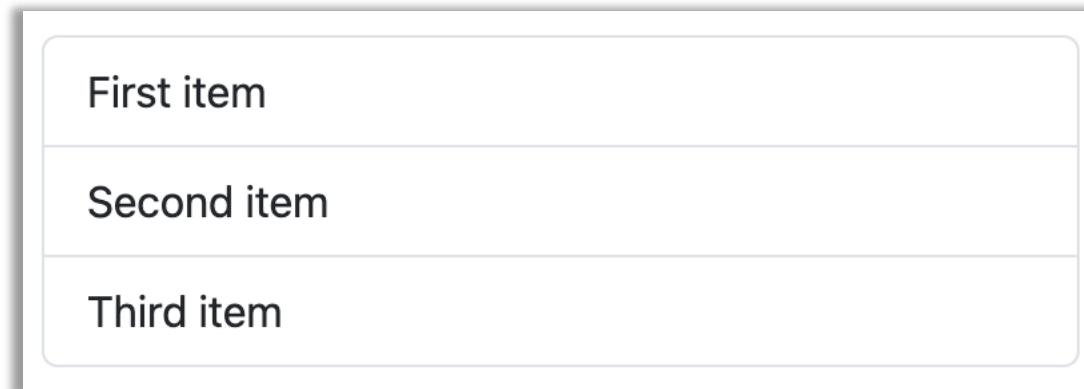
.disabled

.pagination-lg

.pagination-sm

List Group

```
<ul class="list-group">  
  <li class="list-group-item">First item</li>  
  <li class="list-group-item">Second item</li>  
  <li class="list-group-item">Third item</li>  
</ul>
```



Lists with Active Item

.active

```
<ul class="list-group">
  <li class="list-group-item active">First item</li>
  <li class="list-group-item">Second item</li>
  <li class="list-group-item">Third item</li>
</ul>
```

Active item
Second item
Third item

List

- Linked Items
- Disabled Items
- Flush/Remove Borders
- Ordered List Groups
- Horizontal List Groups
- Contextual Classes and link items
- List Groups with Badges

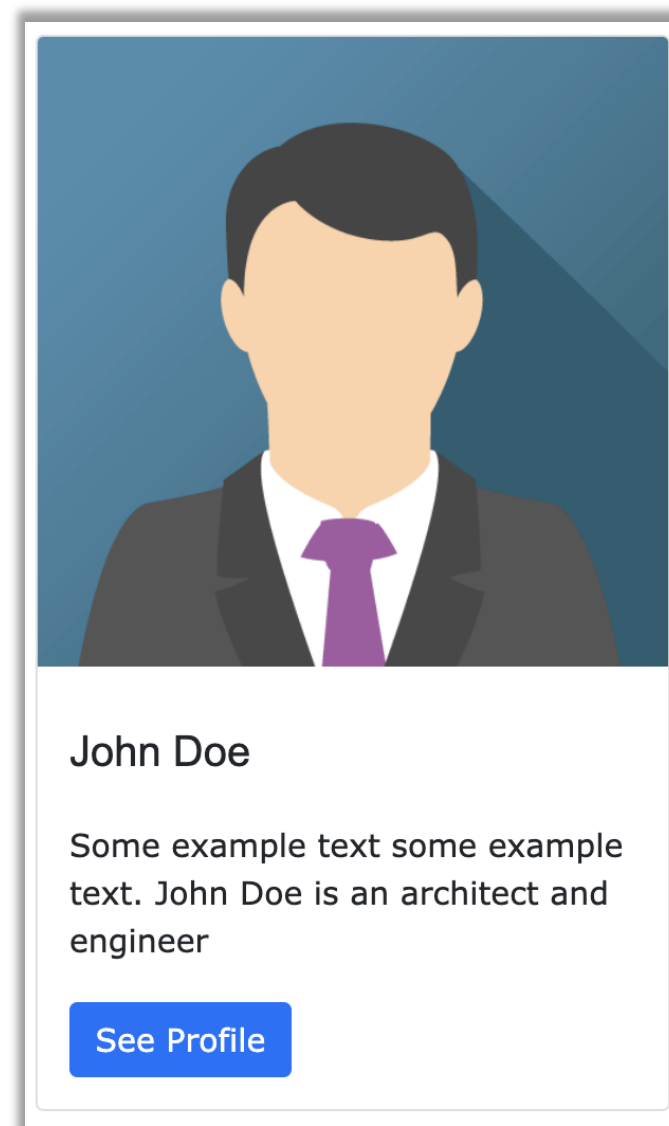
[More details](#)

Cards

- A bordered box with some padding around its content
- Includes options for headers, footers, content, colors, etc.

`.card`

`.card-body`



Cards

```
<div class="card">  
  <div class="card-header">Header</div>  
  <div class="card-body">Content</div>  
  <div class="card-footer">Footer</div>  
</div>
```

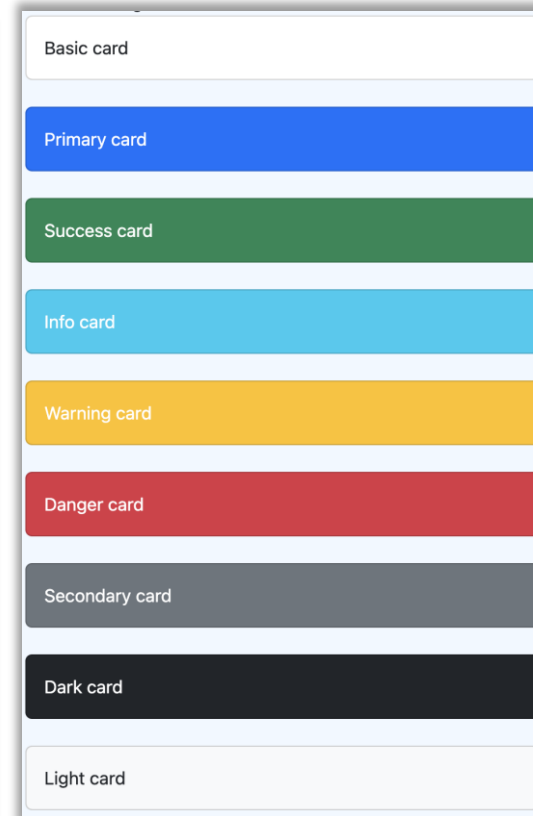
Header

Content

Footer

Contextual Cards

```
<div class="card"><div class="card-body">Basic card</div></div>
<div class="card bg-primary text-white"><div class="card-body">
Primary card</div></div>
<div class="card bg-success text-white"><div class="card-body">
Success card</div></div>
<div class="card bg-info text-white"><div class="card-body">
Info card</div></div>
<div class="card bg-warning text-white"><div class="card-body">
Warning card</div></div>
<div class="card bg-danger text-white"><div class="card-body">
Danger card</div></div>
<div class="card bg-secondary text-white"><div class="card-body">
Secondary card</div></div>
<div class="card bg-dark text-white"><div class="card-body">
Dark card</div></div>
<div class="card bg-light text-dark"><div class="card-body">
Light card</div></div>
```

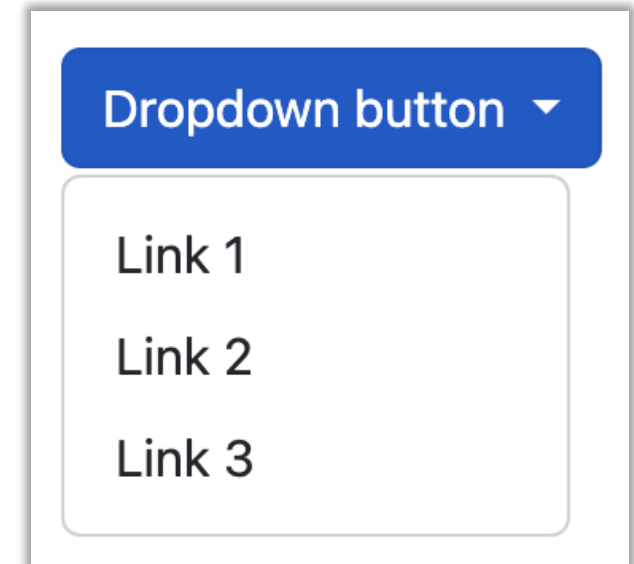


Cards

- Titles, text and links `.card-title .card-text card-link`
- Card Images `.card-img-top .card-img-bottom`
- Card Image Overlays `.card-img-overlay`

Dropdown

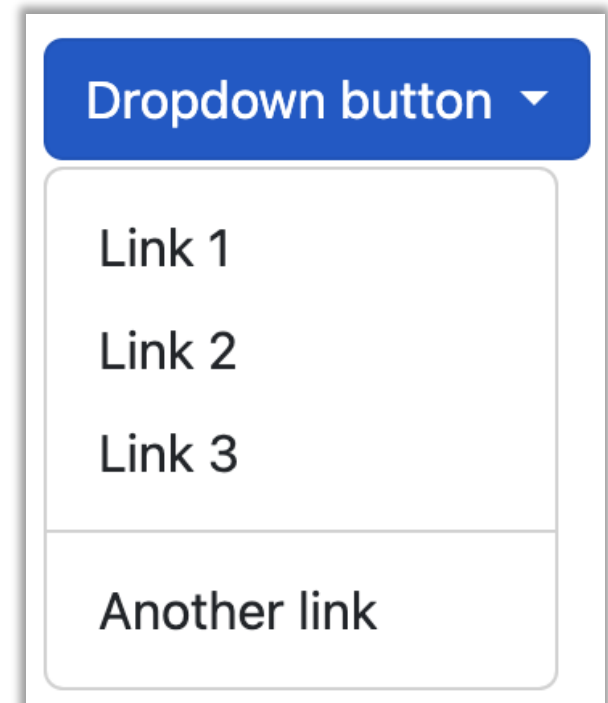
```
<div class="dropdown">
  <button type="button" class="btn btn-primary dropdown-
toggle" data-bs-toggle="dropdown">
    Dropdown button
  </button>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">Link 1</a></li>
    <li><a class="dropdown-item" href="#">Link 2</a></li>
    <li><a class="dropdown-item" href="#">Link 3</a></li>
  </ul>
</div>
```



```
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
```

Dropdown with divider

```
<div class="dropdown">
  <button type="button" class="btn btn-primary dropdown-
toggle" data-bs-toggle="dropdown">
    Dropdown button
  </button>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">Link 1</a></li>
    <li><a class="dropdown-item" href="#">Link 2</a></li>
    <li><a class="dropdown-item" href="#">Link 3</a></li>
    <li><hr class="dropdown-divider"></li>
    <li><a class="dropdown-item" href="#">Another link</a></li>
  </ul>
</div>
```



Dropdown

- Header
- Disable and Active Items
- Position
- Drop Menu, Drop Up
- Text
- Grouped buttons

[More details](#)

Collapsible

- Useful when you want to hide and show large amount of content

```

<button data-bs-toggle="collapse" data-bs-target="#demo"> Collapsible </button>
<div id="demo" class="collapse">
Lorem ipsum dolor text....
</div>
  
```

Collapsible

Lorem ipsum dolor text....

[More details](#)

Nav Menus

Link

Link

Link

Disabled

```
<ul class="nav">
<li class="nav-item">
<a class="nav-link" href="#">Link</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Link</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Link</a>
</li>
<li class="nav-item">
<a class="nav-link disabled" href="#">Disabled</a>
</li>
</ul>
```


Aligned Nav

```
<!-- Centered nav -->  
<ul class="nav justify-content-center">
```

Link Link Link Disabled

Vertical Nav

```
<ul class="nav flex-column">
```

Link

Link

Link

Disabled

Nav Bars

```
<nav class="navbar navbar-expand-sm bg-light">
<div class="container-fluid">
<ul class="navbar-nav">
<li class="nav-item">
<a class="nav-link" href="#">Link 1</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Link 2</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Link 3</a>
</li>
</ul>
</div>
</nav>
```

Link 1 Link 2 Link 3

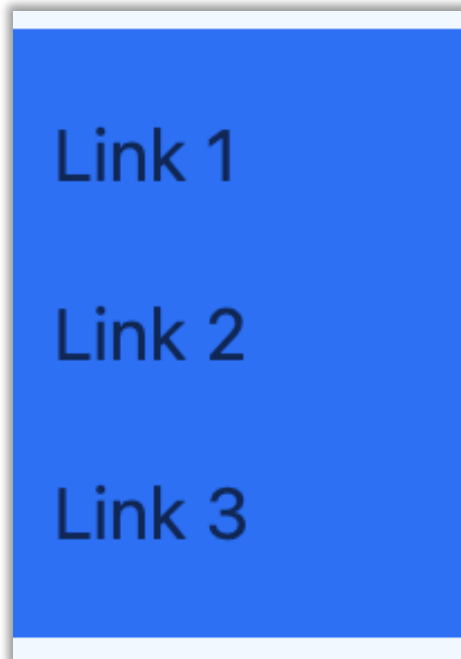
Basic Navbar Example

A navigation bar is a navigation header that is placed at the top of the page.

The navbar-expand-xxl|xl|lg|md|sm class determines when the navbar should stack vertically (on xxlarge, extra large, large, medium or small screens).

Vertical Nav Bar

```
<nav class="navbar bg-primary">
```



Centered Nav Bar

Colored Navbar

```
<nav class="navbar navbar-expand-sm bg-light  
justify-content-center">
```

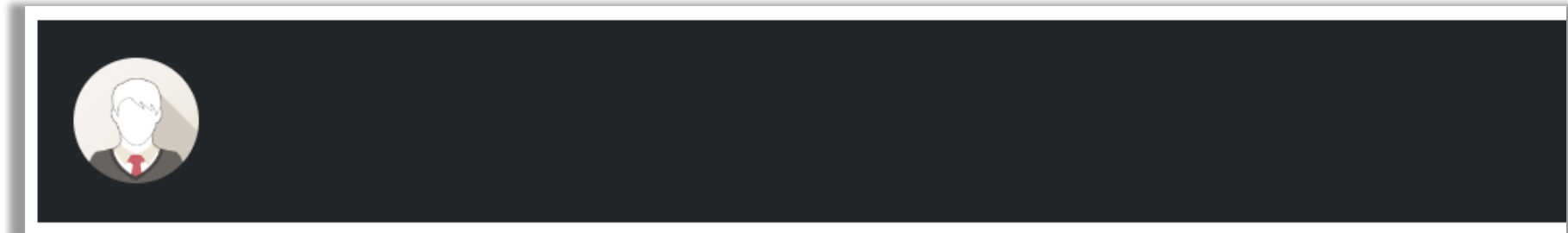
[Link 1](#) [Link 2](#) [Link 3](#)

Brand / Logo

```

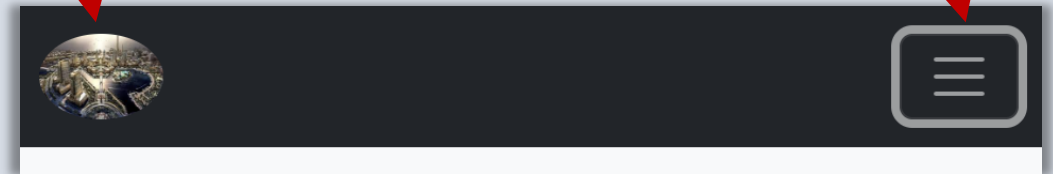
<nav class="navbar navbar-expand-sm bg-dark navbar-dark">
  <div class="container-fluid">
    <a class="navbar-brand" href="#">Logo</a>
  </div>
</nav>

```



Navbar Collapsible

```
<nav class="navbar navbar-expand-sm bg-dark navbar-dark">
  <div class="container-fluid">
    <a class="navbar-brand" href="#"></a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#collapsibleNavbar">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="collapsibleNavbar">
      <ul class="navbar-nav">
        <li class="nav-item">
          <a class="nav-link" href="#">Link</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">Link</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">Link</a>
        </li>
      </ul>
    </div>
  </div>
</nav>
```



Navbar

- Navbar Text
- Navbar with Dropdown
- Navbar Forms and Buttons
- Fixed Nabvar
- Sticky Navbar

[More details](#)

Carousel / Slideshow

- A slideshow for cycling through elements

Class	Description
<code>.carousel</code>	Creates a carousel
<code>.carousel-indicators</code>	Adds indicators for the carousel. These are the little dots at the bottom of each slide (which indicates how many slides there are in the carousel, and which slide the user are currently viewing)
<code>.carousel-inner</code>	Adds slides to the carousel
<code>.carousel-item</code>	Specifies the content of each slide
<code>.carousel-control-prev</code>	Adds a left (previous) button to the carousel, which allows the user to go back between the slides
<code>.carousel-control-next</code>	Adds a right (next) button to the carousel, which allows the user to go forward between the slides
<code>.carousel-control-prev-icon</code>	Used together with <code>.carousel-control-prev</code> to create a "previous" button
<code>.carousel-control-next-icon</code>	Used together with <code>.carousel-control-next</code> to create a "next" button
<code>.slide</code>	Adds a CSS transition and animation effect when sliding from one item to the next. Remove this class if you do not want this effect



```
<div id="demo2" class="carousel slide" data-bs-ride="carousel">
<div class="carousel-indicators">
<button type="button" data-bs-target="#demo2" data-bs-slide-to="0" class="active" aria-current="true"></button>
<button type="button" data-bs-target="#demo2" data-bs-slide-to="1" class=""></button>
<button type="button" data-bs-target="#demo2" data-bs-slide-to="2" class=""></button>
</div>
<div class="carousel-inner"> <div class="carousel-item active">

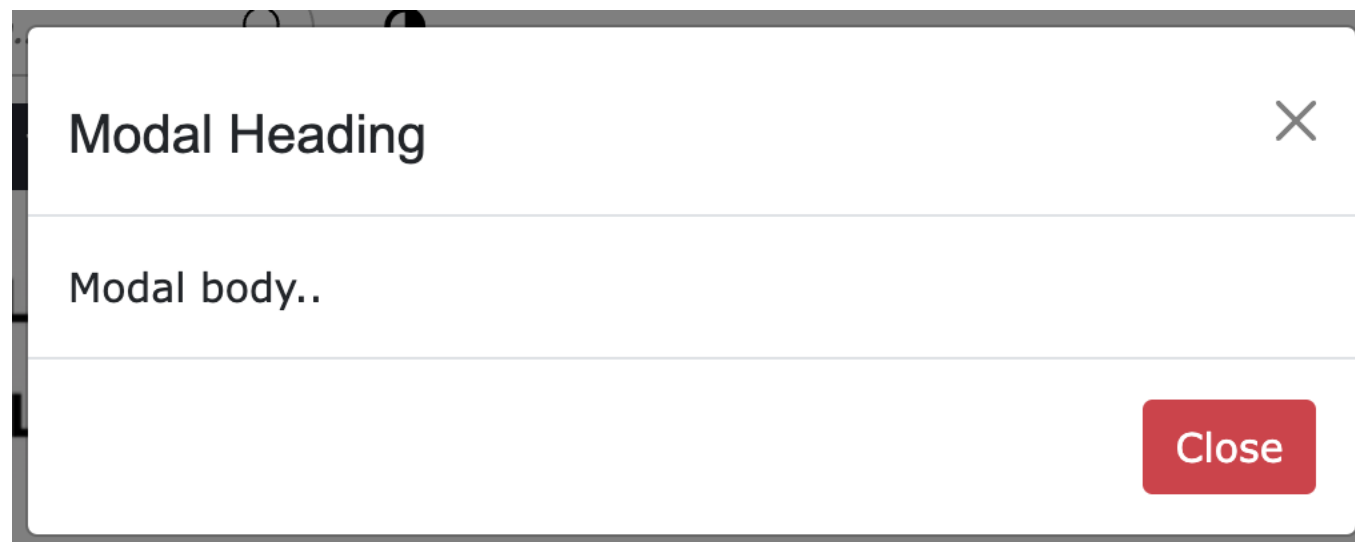
<div class="carousel-caption"> <h3>Los Angeles</h3> <p>We had such a great time in LA!</p></div></div>
<div class="carousel-item"><div class="carousel-
caption"><h3>Chicago</h3><p>Thank you, Chicago!</p></div> </div>
<div class="carousel-item"><div class="carousel-
caption"><h3>New York</h3><p>We love the Big Apple!</p> </div> </div> </div>
<button class="carousel-control-prev" type="button" data-bs-target="#demo2" data-bs-slide="prev"
style="cursor:pointer"><span class="carousel-control-prev-icon"></span>
</button><button class="carousel-control-next" type="button" data-bs-target="#demo2" data-bs-slide="next"
style="cursor:pointer"><span class="carousel-control-next-icon"></span></button></div>
```



[More details](#)

Modals

- Dialog or Popup
- Display on top of the current page



Modals - Dialog Box / Popup

```
<!-- Button to Open the Modal -->
<button type="button" class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#myModal">
  Open modal
</button>
<!-- The Modal -->
<div class="modal" id="myModal">
  <div class="modal-dialog">
    <div class="modal-content">

      <!-- Modal Header -->
      <div class="modal-header">
        <h4 class="modal-title">Modal Heading</h4>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <!-- Modal body -->
      <div class="modal-body">
        Modal body..
      </div>
      <!-- Modal footer -->
      <div class="modal-footer">
        <button type="button" class="btn btn-danger" data-bs-dismiss="modal">Close</button>
      </div>
    </div>
  </div>
</div>
```

```
<!-- Fading modal -->
<div class="modal fade"></div>
```

Modal Size

`.modal-sm`

`.modal-lg`

`.modal-xl`

`.modal-fullscreen`

Responsive Fullscreen Modal

Class	Description
<code>.modal-fullscreen-sm-down</code>	Fullscreen below 576px
<code>.modal-fullscreen-md-down</code>	Fullscreen below 768px
<code>.modal-fullscreen-lg-down</code>	Fullscreen below 992px
<code>.modal-fullscreen-xl-down</code>	Fullscreen below 1200px
<code>.modal-fullscreen-xxl-down</code>	Fullscreen below 1400px

Modal

- Centered Modal

`.modal-dialog-centered`

- Scrolling Modal

```
<div class="modal-dialog modal-dialog-scrollable">
```


Tooltip

The Tooltip component

Hooray!

Hover over me!

- Tooltip
- Tooltip position

`data-bs-placement`

```
<a href="#" data-bs-toggle="tooltip" data-bs-placement="top" title="Hooray!">Hover</a>
<a href="#" data-bs-toggle="tooltip" data-bs-placement="bottom" title="Hooray!">Hover</a>
<a href="#" data-bs-toggle="tooltip" data-bs-placement="left" title="Hooray!">Hover</a>
<a href="#" data-bs-toggle="tooltip" data-bs-placement="right" title="Hooray!">Hover</a>
```

More Bootstrap features

- **Popover**: Pop-up box on click
 - Position, closing, Hoverable - [more](#)
- **Toast**: an alert box that is only shown for a couple of seconds when something happens
[more](#)
- **Scrollspy**: automatically update links in a navigation list based on scroll position [more](#)

More Bootstrap features

- **Offcanvas**: hidden modal - used as a sidebar navigation menu [[more](#)]
- [Utilities](#)
- [Dark mode](#)
- [Flex](#)

CSS Animations

- Allows animation of HTML elements without using JavaScript
- Let's an element gradually change from one style to another
- Specify some **keyframes** for the animation
 - Hold what styles the element will have at certain times

CSS Animation

- How to use

```
/* The animation code */
@keyframes example {
  from {background-color: red;}
  to {background-color: yellow;}
}
```

```
/* The element to apply the animation to */
div {
  width: 100px;
  height: 100px;
  background-color: red;
  animation-name: example;
  animation-duration: 4s;
}
```

CSS Animation

```
/* The animation code */  
@keyframes example {  
  0%    {background-color: red;}  
  25%   {background-color: yellow;}  
  50%   {background-color: blue;}  
  100%  {background-color: green;}  
}
```

CSS Animation

```
/* The animation code */  
@keyframes example {  
  0%    {background-color:red; left:0px; top:0px;}  
  25%   {background-color:yellow; left:200px; top:0px;}  
  50%   {background-color:blue; left:200px; top:200px;}  
  75%   {background-color:green; left:0px; top:200px;}  
  100%  {background-color:red; left:0px; top:0px;}  
}
```

CSS Animation - Delay

- **animation-delay** : Delay for the start of an animation

```
div {  
  width: 100px;  
  height: 100px;  
  position: relative;  
  background-color: red;  
  animation-name: example;  
  animation-duration: 4s;  
  animation-delay: 2s;  
}
```


CSS Animation - Count

- **animation-iteration-count** : number of times an animation should run

```
div {  
  width: 100px;  
  height: 100px;  
  position: relative;  
  background-color: red;  
  animation-name: example;  
  animation-duration: 4s;  
  animation-delay: 2s;  
  animation-iteration-count: 3;  
}
```

CSS Animation - Direction

- **animation-direction** : play forward, backwards
- **normal** - forwards
- **reverse** - backwards
- **alternate** - forwards first, then backwards
- **alternate-reverse** - backwards first then forward

```
div {  
    ...  
    animation-direction: reverse;  
}
```

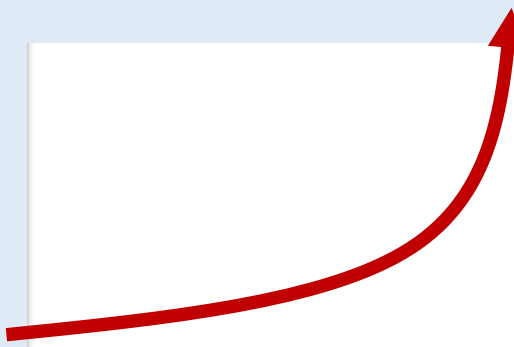
CSS Animation

- Specify the speed of movements `animation-timing-function`
- Specify the fill-mode `animation-fill-mode`

```
div {
  animation-name: example;
  animation-duration: 5s;
  animation-timing-function: linear;
  animation-delay: 2s;
  animation-iteration-count: infinite;
  animation-direction: alternate;
}

div {
  animation: example 5s linear 2s infinite alternate;
}
```

Shorthand



CSS Transform

```
<!DOCTYPE html>
<html>
<head>
<style>
div.a {
width: 150px;
height: 80px;
background-color: yellow;
transform: rotate(20deg);
}

div.b {
width: 150px;
height: 80px;
background-color: yellow;
transform: skewY(20deg);
}

div.c {
width: 150px;
height: 80px;
background-color: yellow;
transform: scaleY(1.5);
}
</style>
</head>
<body>

<h1>The transform Property</h1>

<h2>transform: rotate(20deg):</h2>
<div class="a">Hello World!</div>
<br>

<h2>transform: skewY(20deg):</h2>
<div class="b">Hello World!</div>
<br>

<h2>transform: scaleY(1.5):</h2>
<div class="c">Hello World!</div>

</body>
</html>
```

The transform Property

transform: rotate(20deg):



transform: skewY(20deg):



transform: scaleY(1.5):



CSS Transform

The transform-origin Property

```
<html>
<head>
<style>
#div1 {
  position: relative;
  height: 200px;
  width: 200px;
  margin: 50px;
  padding: 10px;
  border: 1px solid black;
}

#div2 {
  padding: 50px;
  position: absolute;
  border: 1px solid black;
  background-color: red;
  transform: rotate(45deg);
}

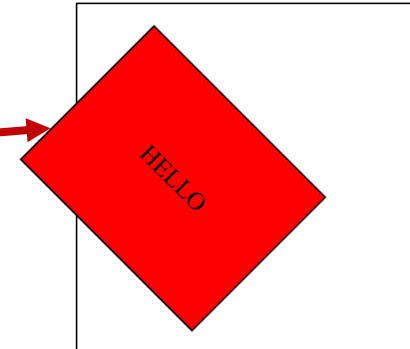
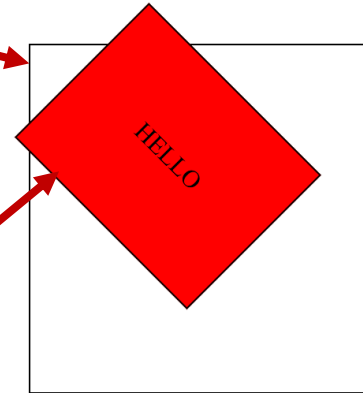
#div3 {
  padding: 50px;
  position: absolute;
  border: 1px solid black;
  background-color: red;
  transform: rotate(45deg);
  transform-origin: 10% 40%;
}
</style>
</head>
<body>

<h1>The transform-origin
Property</h1>

<div id="div1">
  <div id="div2">HELLO</div>
</div>

<div id="div1">
  <div id="div3">HELLO</div>
</div>

</body>
</html>
```



```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  position: relative;
  width: 100px;
  height: 100px;
  background-color: red;
  animation-name: myAnimation;
  animation-duration: 10s;
}

@keyframes myAnimation {
  0% {background-color:red; transform: rotate(360deg);}
  25% {background-color:yellow;}
  50% {background-color:blue;}
  75% {background-color:green;}
  100% {background-color:red;}
}
</style>
</head>
<body>

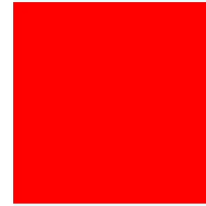
<h1>CSS Animation</h1>

<div></div>

<p><b>Note:</b> When an animation is finished, it goes
back to its original style.</p>

</body>
</html>
```

CSS Animation



Note: When an animation is finished, it goes back to its original style.

Forms

Stacked Form

- All textual `<input>` and `<textarea>` elements with class `.form-control` get proper form styling

```
<form action="/action_page.php">
  <div class="mb-3 mt-3">
    <label for="email" class="form-label">Email:</label>
    <input type="email" class="form-control" id="email" placeholder="Enter email" name="email">
  </div>
  <div class="mb-3">
    <label for="pwd" class="form-label">Password:</label>
    <input type="password" class="form-control" id="pwd" placeholder="Enter password" name="pswd">
  </div>
  <div class="form-check mb-3">
    <label class="form-check-label">
      <input class="form-check-input" type="checkbox" name="remember"> Remember me
    </label>
  </div>
  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

Email:

Password:

☐ Remember me

Textarea

```

<label for="comment">Comments:</label>
<textarea class="form-control" rows="5" id="comment" name="text"></textarea>
  
```

Comments:

Submit

Form Row/Grid - Inline Forms

```
<form>
  <div class="row">
    <div class="col">
      <input type="text" class="form-control" placeholder="Enter email" name="email">
    </div>
    <div class="col">
      <input type="password" class="form-control" placeholder="Enter password" name="pswd">
    </div>
  </div>
</form>
```

From Control Size

```
<input type="text" class="form-control form-control-lg" placeholder="Large input">
<input type="text" class="form-control" placeholder="Normal input">
<input type="text" class="form-control form-control-sm" placeholder="Small input">
```

Large input

Normal input

Small input

Readonly and Disabled

```
<input type="text" class="form-control" placeholder="Normal input">  
<input type="text" class="form-control" placeholder="Disabled input" disabled>  
<input type="text" class="form-control" placeholder="Readonly input" readonly>
```

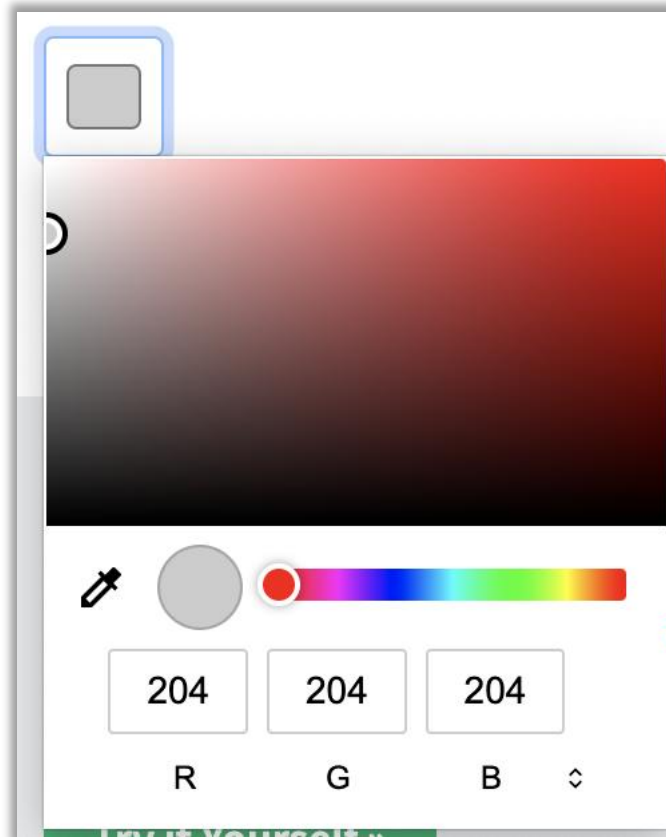
Normal input

Disabled input

Readonly input

Date Picker

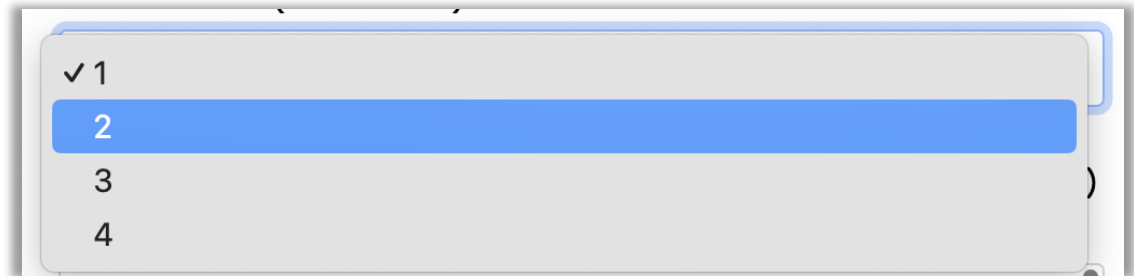
```
<input type="color" class="form-control form-control-color" value="#CCCCCC">
```



Select Menu

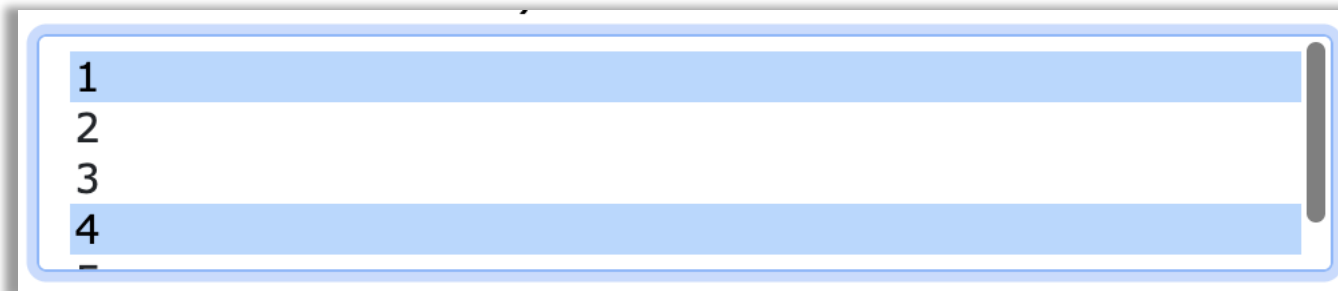
`.form-select-lg` `.form-select-sm`

```
<select class="form-select">
  <option>1</option>
  <option>2</option>
  <option>3</option>
  <option>4</option>
</select>
```



```
<select class="form-select" multiple>
```

Multiple Select



[More details](#)

Form Validation

```
<form class="was-validated">
```

```
<div class="mb-3 mt-3">
  <label for="uname" class="form-label">Username:</label>
  <input type="text" class="form-control" id="uname" placeholder="Enter
username" name="uname" required>
  <div class="valid-feedback">Valid.</div>
  <div class="invalid-feedback">Please fill out this field.</div>
</div>
```

Username:

Enter username



Please fill out this field.

Username:

abbas



Valid.

More Forms Controls

- Checkboxes, Radio and Toggle buttons
- Range
- Input Groups
- Floating Labels

More Bootstrap

- Grids
- Templates

Reference

- <https://www.w3schools.com/bootstrap5/index.php>
- <https://mdbootstrap.com/docs/standard/components/button-group/>